3c Iterative Regressions

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Imports

From code "3b Preliminary Results:" To improve this regression,

1. DONE. ~~I want to exclude outliers. So, drop organizations that have multiple years of zero program expenses and organizations that are so large they even skew log(revenues) and may not need to respond at all the economic conditions.~~ 1b. I want to exclude outliers. So, drop schools and hospitals that are not a type of organization that fluctuates directly with economic conditions (demand is more inelastic).
2. NOT POSSIBLE. ~~I want to exclude multi-state non-profits. So, a subset of the the data which would be limited to only local non-profit organizations that do activities within their own state and would be more closely responding to the local conditions instead of the needs of the country or multiple states.~~
3. DONE. ~~I want to try excluding non-profits that did not exist in the Covid-19 recession. So, a subset of the data which would be limited only to non-profits that responded (in any extent) to an economic recession.~~
4. DONE. ~~Further, I want to try a subset of only organizations that expanded during Covid-19, to see if my variables interact with them differently. I may do this through a Boolean, interaction variable.~~
5. Double check that Program Expense variable includes grants to individuals/other orgs/governments. Might be different variables, in that case I will have to combine them.
6. Diagnose why so many observations get removed because of "NA."

```r
org_data = readRDS("../Final_Data/clean_master_data.rds")

#Packages
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(fixest)
```

Regression 1.0

Same as Regression 1.0 from code 3b

```r
#Setup code
regression_variables = org_data %>%
  select(F9_09_EXP_TOT_PROG, gdp_change_percent, YEAR, F9_00_ORG_ADDR_STATE, activity_category, F9_08_RE

  rename( #renaming to match all conventions and make them easier to understand
    total_program_expenses = F9_09_EXP_TOT_PROG,
    year = YEAR,
    state = F9_00_ORG_ADDR_STATE,
    industry = activity_category,
    total_revenue = F9_08_REV_TOT_TOT,
    organization_ein = ORG_EIN
  ) %>%

  mutate(
    log_program_expenses = log(total_program_expenses + 1),
    industry = toupper(trimws(industry)),
    log_revenue = log(total_revenue + 1)
    )
baseline_model = feols(log_program_expenses ~ gdp_change_percent + log_revenue | organization_ein + year
```

```
## NOTES: 544,862 observations removed because of NA values (Fixed-effects: 544,862).
##        65,647/0/0/0 fixed-effect singletons were removed (65,647 observations).
```

```r
summary(baseline_model)
```

```
## OLS estimation, Dep. Var.: log_program_expenses
## Observations: 4,392,247
## Fixed-effects: organization_ein: 548,897,  year: 15,  state: 50,
##  industry: 26
## Standard-errors: IID
##                    Estimate Std. Error    t value  Pr(>|t|)
## gdp_change_percent 0.000936   0.000568    1.64904   0.09914 .
## log_revenue        0.846574   0.000331 2555.45787 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 1.47635     Adj. R2: 0.943563
##                 Within R2: 0.629521
```

1.1 Exclude Outliers

Removing organizations with multiple years of zero program expenses and extremely large organizations:

```r
# Identify organizations with multiple years of zero program expenses
orgs_with_zeros = regression_variables %>%
  group_by(organization_ein) %>%
  summarize(years_with_zero = sum(total_program_expenses == 0, na.rm = TRUE)) %>%
  filter(years_with_zero >= 2) %>%
  pull(organization_ein)

# Remove outliers based on log_revenue (top 1% largest orgs)
revenue_threshold = quantile(regression_variables$log_revenue, 0.99, na.rm = TRUE)

# Drop schools and churches
```

```r
# Create filtered data set
regression_no_outliers = regression_variables %>%
  filter(!organization_ein %in% orgs_with_zeros, #exclude zero expense orgs
         log_revenue <= revenue_threshold, #exclude top 1% revenue orgs
         org_type == "Neither" #exclude schools and churches
         )

# Run regression without outliers
model_no_outliers = feols(log_program_expenses ~ gdp_change_percent + log_revenue |
                           organization_ein + state + year + industry,
                         data = regression_no_outliers)
```

```
## NOTES: 48 observations removed because of NA values (Fixed-effects: 48).
##        62,708/0/0/0 fixed-effect singletons were removed (62,708 observations).
```

```r
summary(model_no_outliers)
```

```
## OLS estimation, Dep. Var.: log_program_expenses
## Observations: 1,989,095
## Fixed-effects: organization_ein: 227,099,  state: 50,  year: 15,
##  industry: 26
## Standard-errors: IID
##                     Estimate Std. Error    t value  Pr(>|t|)
## gdp_change_percent -0.000614   0.000487   -1.26093   0.20733
## log_revenue         0.842345   0.000470 1793.03740 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.869557     Adj. R2: 0.860299
##                  Within R2: 0.645985
```

1.2 Exclude Multi-State Non-Profits

Upon thorough review, this is impossible to do with existing data sets. However, by removing large organizations, I am increasingly likely to limit the set to smaller, local organizations.

1.3 Existed in COVID-19

Limiting to organizations that were active during the COVID-19 recession (2020):

```r
# Identify organizations that existed in 2020
covid_orgs = regression_no_outliers %>%
  filter(year == 2020) %>%
  pull(organization_ein) %>%
  unique()

# Create COVID-era dataset
regression_covid_orgs = regression_no_outliers %>%
  filter(organization_ein %in% covid_orgs)

# Run regression for COVID-era organizations
model_covid_orgs = feols(log_program_expenses ~ gdp_change_percent + log_revenue |
                          organization_ein + state + year + industry,
                        data = regression_covid_orgs)
```

```
## NOTES: 48 observations removed because of NA values (Fixed-effects: 48).
##        2,652/0/0/0 fixed-effect singletons were removed (2,652 observations).
```

```r
summary(model_covid_orgs)
```

```
## OLS estimation, Dep. Var.: log_program_expenses
## Observations: 1,888,610
## Fixed-effects: organization_ein: 200,671,  state: 50,  year: 15,
##  industry: 26
## Standard-errors: IID
##                     Estimate Std. Error     t value  Pr(>|t|)
## gdp_change_percent -0.000456   0.000476   -0.959304   0.33741
## log_revenue         0.833939   0.000498 1675.889306 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.834172     Adj. R2: 0.855407
##                  Within R2: 0.624631
```

1.4 Expanded in COVID-19

Creating an indicator for organizations that expanded during COVID and testing interaction effects:

```r
# Identify organizations that expanded during COVID (2019-2020). Thanks, AI.
covid_expansion = regression_covid_orgs %>%
  filter(year %in% c(2019, 2020)) %>%
  arrange(organization_ein, year) %>%
  group_by(organization_ein) %>%
  summarize(
    expanded_covid = if_else(
      n() == 2 &
        total_program_expenses[year == 2020] > total_program_expenses[year == 2019],
      1, 0
    )
  )
```

```
## Warning: Returning more (or less) than 1 row per `summarise()` group was deprecated in
## dplyr 1.1.0.
## i Please use `reframe()` instead.
## i When switching from `summarise()` to `reframe()`, remember that `reframe()`
##   always returns an ungrouped data frame and adjust accordingly.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `summarise()` has grouped output by 'organization_ein'. You can override using
## the `.groups` argument.
```

```r
# Merge expansion indicator back to data
regression_with_expansion = regression_covid_orgs %>%
  left_join(covid_expansion, by = "organization_ein") %>%
  mutate(expanded_covid = if_else(is.na(expanded_covid), 0, expanded_covid))

# Run regression with interaction term
model_expansion_interaction = feols(
  log_program_expenses ~ gdp_change_percent * expanded_covid + log_revenue | organization_ein + state +
  data = regression_with_expansion
)
```

```
## NOTES: 48 observations removed because of NA values (Fixed-effects: 48).
##        2,652/0/0/0 fixed-effect singletons were removed (2,652 observations).
```

```r
summary(model_expansion_interaction)
```

```
## OLS estimation, Dep. Var.: log_program_expenses
## Observations: 1,888,610
## Fixed-effects: organization_ein: 200,671,  state: 50,  year: 15,
##  industry: 26
## Standard-errors: IID
##                                   Estimate  Std. Error       t value  Pr(>|t|)
## gdp_change_percent                 0.006950    0.000529   13.14659312 < 2.2e-16
## expanded_covid                    -0.021804 4916.804049   -0.00000443         1
## log_revenue                        0.833355    0.000498 1674.10990847 < 2.2e-16
## gdp_change_percent:expanded_covid -0.017822    0.000555  -32.09620963 < 2.2e-16
##
## gdp_change_percent                ***
## expanded_covid
## log_revenue                       ***
## gdp_change_percent:expanded_covid ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.833918     Adj. R2: 0.855495
##                  Within R2: 0.62486
```

Next Steps?